

Consumer Choice and the Demand Curve

Martin J. Conyon

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Required readings

Mankiw, *Principles of Microeconomics*

Chapter 21 — The Theory of Consumer Choice

Budget constraints, indifference curves, the consumer's optimum, and how demand curves follow from it.

One difference worth knowing. Mankiw works the chapter without calculus. These slides use it — marginal utilities as derivatives, the optimum by Lagrange — so what follows is the more general treatment, and the chapter is the gentler route into the same result.

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Slide 2

Working through it

What the chapter gives you. Mankiw chapter 21 builds the same apparatus in the same order: the budget constraint first, then indifference curves, then the point where they touch, then the derivation of a demand curve.

Where these slides differ, and why it matters. Mankiw states the tangency condition and asks you to accept the marginal rate of substitution as the slope of an indifference curve.

Here it is *derived*: the marginal utilities are partial derivatives, their ratio is the rate of substitution, and the optimum falls out of a Lagrangian. Nothing has to be taken on trust.

The practical consequence. Everything in the chapter is in this deck, plus the machinery that proves it. Read the chapter first if the algebra here moves quickly — the words come easier when you have already met the picture.

Where we are going

Where we are going

1. Preferences and indifference curves
2. The budget constraint
3. The optimum — argued before it is solved
4. Solving it
5. A price change
6. Deriving the demand curve
7. Income and substitution effects
8. Why the substitution effect is never positive
9. What the model leaves out

Every number in the lecture comes from one utility function and four prices.

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Slide 3

Working through it

The shape of the argument. Parts 1 and 2 build the two halves of the problem separately: what the consumer *wants* and what the consumer *can afford*. Part 3 puts them together and finds the best affordable bundle by argument and by arithmetic. Part 4 does the same thing with calculus and gets a formula.

Parts 5 and 6 are where the payoff is. Change one price, re-solve, and you have a second point. Change it again and again and you have a curve — and that curve is the demand curve, which up to now has been an assumption.

Parts 7 and 8 take the response apart. A price rise does two things at once, and separating them is what lets us prove something that is otherwise only asserted: that a compensated demand curve can never slope upward.

One utility function throughout. $U(x, y) = \sqrt{xy}$, income \$120 a week, $p_y = \$2$, and p_x moving between \$3 and \$6. Every figure and every table in the deck is computed from those.

1. Part 1: Preferences and Indifference Curves

A bundle, and what we assume about it

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Bundle. A list of quantities — so much of one good, so much of another. With two goods it is a pair of numbers, (x, y) .

Three assumptions, and they are all we will use:

1. **More is better.** A bundle with more of both is preferred.
2. **Completeness.** Any two bundles can be compared.
3. **Consistency.** If A is preferred to B and B to C , then A is preferred to C .

Nothing else is assumed about taste.

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Slide 5

Working through it

Why start here. A theory of choice needs something to choose between and some discipline on how the choosing works. The bundle supplies the first. The three assumptions supply the second, and they are deliberately weak.

What each assumption does.

- **More is better** makes the problem interesting: without it the consumer might not spend the budget at all.
- **Completeness** rules out “I cannot say” — the consumer always has an answer.
- **Consistency** (transitivity) rules out preference cycles. It is the assumption doing the real work in Part 8, where it is the only thing the proof uses.

What is deliberately absent. No assumption that the consumer is selfish, well-informed, sensible, or right. Preferences are whatever they are. The model takes them as data and asks only what follows.

The utility function we will use

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Utility. A number attached to a bundle, arranged so that a preferred bundle carries a higher number. Only the *ordering* means anything.

Throughout:

$$U(x, y) = x^{1/2} y^{1/2} = \sqrt{xy}$$

Feed it a bundle and it returns a number:

$$U(20, 30) = \sqrt{600} \approx 24.5, \quad U(10, 30) = \sqrt{300} \approx 17.3$$

The size of the number means nothing. That $24.5 > 17.3$ means everything.

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Slide 6

Working through it

What a utility function is and is not. It is a bookkeeping device. It converts a ranking of bundles into numbers so the ranking can be worked with algebraically. It is *not* a measure of pleasure, happiness or anything psychological, and no claim is made that 24.5 is “1.4 times as good” as 17.3.

Why the ordering is all that matters. Any strictly increasing transformation of U ranks bundles identically. That is why we may compare xy instead of $\sqrt{xy}$ whenever it is convenient — squaring is strictly increasing on positive numbers, so it preserves the ranking exactly. Part 3 uses this to avoid square roots in a table.

The two numbers checked. $20 \times 30 = 600$ and $\sqrt{600} = 24.49 \dots$ $10 \times 30 = 300$ and $\sqrt{300} = 17.32 \dots$

Why this function. It has a name — Cobb–Douglas — and it is chosen because everything it produces is exact and whole. Part 9 says what that convenience costs.

Step 1 — write the indifference curve as an equation**Step 1 — write the indifference curve as an equation**

Hold utility fixed at some level and ask which bundles deliver it.

$$\sqrt{xy} = \text{constant} \quad \implies \quad xy = k \quad \implies \quad y = \frac{k}{x}$$

Squaring both sides removes the root and costs nothing, because squaring does not disturb the ranking.

Through the bundle (20, 30): $k = 20 \times 30 = 600$, so

$$y = \frac{600}{x}$$

An indifference curve is the graph of an equation, not a shape drawn by feel.

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Slide 7

Working through it

Definition. An **indifference curve** is the set of bundles among which the consumer is indifferent — every point on it delivers the same utility.

The algebra, one step at a time.

$$\sqrt{xy} = c \implies xy = c^2 \implies y = \frac{c^2}{x}$$

Write $k = c^2$ and the curve is $y = k/x$: a rectangular hyperbola with a known constant.

Why squaring is legitimate. Squaring is strictly increasing on positive numbers, so $\sqrt{x_1 y_1} > \sqrt{x_2 y_2}$ exactly when $x_1 y_1 > x_2 y_2$. The ranking survives untouched, and the algebra gets much easier.

Which curve you get. One for every value of k — a whole family filling the positive quadrant. Larger k means a curve further from the origin and a higher utility. The curve through a *particular* bundle is found by multiplying its two coordinates: $k = x_0 y_0$.

Step 2 — plug the numbers in**Step 2 — plug the numbers in**

Every row divides 600 by x . Read across and multiply: the answer is 600 every time.

x	10	12	15	20	30	40	60
$y = 600/x$	60	50	40	30	20	15	10
xy	600	600	600	600	600	600	600

The consumer is **indifferent** between all seven. Each is $U = \sqrt{600}$.

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Slide 8

Working through it

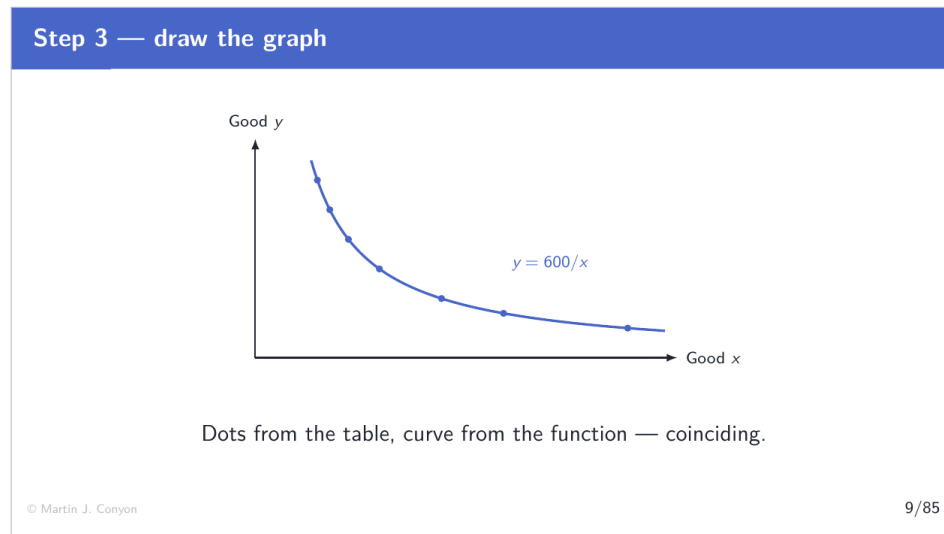
Every entry, checked.

$$\begin{array}{llll}
 10 \times 60 = 600 & 12 \times 50 = 600 & 15 \times 40 = 600 & 20 \times 30 = 600 \\
 30 \times 20 = 600 & 40 \times 15 = 600 & 60 \times 10 = 600 &
 \end{array}$$

What to notice in the table. The values are chosen so that both coordinates are whole numbers, and the list is symmetric: (10,60) and (60,10) are both on the curve, as are (12,50) and (50,12). That symmetry is a feature of $xy = k$ — swapping x and y leaves the product alone.

The economic content. The consumer would trade (10,60) for (20,30) without hesitating and without gaining. Fifty units of y given up for ten units of x is, to this consumer at these quantities, an even swap.

What the table is not. It is seven samples of a continuous curve. Every point between the samples is on it too.

Step 3 — draw the graph

Slide 9

Working through it

How the figure is built. The curve is plot of $y = k/x$ with $k = \backslash k_A = 600$, computed from the declared bundle $(x_A, y_A) = (20, 30)$. The dots are placed by a loop that computes $y = 600/x$ for each tabulated x . No coordinate is typed in.

The visual check. If the dots did not land on the curve, either the table or the function would be wrong. They land on it because both come from the same k .

Two properties visible in the shape.

- **Downward sloping.** More x with the same utility requires less y . This follows from “more is better”: if you could have more of both and stay indifferent, more would not be better.
- **Convex to the origin.** The curve flattens as x rises. At the left the consumer has little x and much y , and will part with a lot of y for a unit of x . At the right the position is reversed.

The asymptotes. $y = 600/x$ never reaches either axis. The consumer will not give up the last unit of either good at any rate of exchange — a property of this function, and a mild one.

Two indifference curves cannot cross

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Suppose two curves crossed at a bundle P .

- P is on the lower curve, so $U(P)$ is the lower utility.
- P is on the higher curve, so $U(P)$ is the higher utility.

One bundle, two different utilities. Impossible.

The argument uses nothing but consistency — assumption 3. It is the first time that assumption earns its keep, and not the last.

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Slide 10

Working through it

The argument in full. Let curve I_1 carry utility u_1 and curve I_2 carry utility u_2 , with $u_1 \neq u_2$. Suppose both pass through the bundle P . Since $P \in I_1$, $U(P) = u_1$. Since $P \in I_2$, $U(P) = u_2$. So $u_1 = u_2$, contradicting the assumption that the curves carry different utilities.

Why it matters. It means the indifference map is a clean set of nested curves, and “higher curve” unambiguously means “better”. Without it, the phrase “the highest curve the consumer can reach” — which is the whole statement of the problem in Part 3 — would not be meaningful.

The general form of the argument. A bundle cannot be strictly better than itself. Every result in this deck that looks like a contradiction has this shape, including the substitution-effect proof in Part 8, which contradicts itself in exactly the same way: A preferred to C and C preferred to A .

A common confusion. Curves may come arbitrarily close without crossing. Nothing here rules that out; only intersection is ruled out.

The slope: the marginal rate of substitution

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Marginal rate of substitution. The amount of y the consumer will give up for one more unit of x , with utility held constant. It is the slope of the indifference curve, without its minus sign.

From $\rightarrow$ to	Δy	Δx	chord slope
$(10, 60) \rightarrow (20, 30)$	-30	+10	-3
$(20, 30) \rightarrow (40, 15)$	-15	+20	-0.75

The curve is flattening: a *diminishing rate of substitution*.

At a point the exact value is $MRS = y/x$, proved in Part 4.

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Working through it

The two chords, computed.

$$\frac{30 - 60}{20 - 10} = \frac{-30}{10} = -3, \quad \frac{15 - 30}{40 - 20} = \frac{-15}{20} = -0.75$$

Why the rate diminishes. At $(10, 60)$ the consumer is awash in y and short of x , so an extra x is worth a great deal of y . At $(40, 15)$ the position is reversed. Diminishing marginal rate of substitution is the formal statement of a familiar idea: the more you have of something, the less another unit is worth to you *in terms of what you would give up for it*.

Chords versus the exact slope. A chord is the average rate over an interval. The MRS proper is the rate at a single point. On $y = 600/x$ the point values at the ends of the first chord are

$$\left. \frac{y}{x} \right|_{(10,60)} = 6, \quad \left. \frac{y}{x} \right|_{(20,30)} = 1.5$$

and the chord slope of -3 lies between them, as an average must.

Where $MRS = y/x$ comes from. Part 4 differentiates U and takes the ratio of the two marginal utilities. Everything here is arithmetic that the calculus will confirm.

2. Part 2: The Budget Constraint

Step 1 — write the budget constraint

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Preferences say what the consumer *wants*. The budget says what they *can have*.

Income \$120 a week. $p_x = \$3$. $p_y = \$2$.

Spending on x , plus spending on y , equals income:

$$\underbrace{3x}_{\text{spent on } x} + \underbrace{2y}_{\text{spent on } y} = \underbrace{120}_{\text{income}}$$

Every coefficient is a price. The right-hand side is income. Nothing here refers to taste.

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Slide 15

Working through it

The general form. With two goods, prices p_x and p_y and income m ,

$$p_x x + p_y y = m$$

Here $p_x = 3$, $p_y = 2$, $m = 120$.

Why it is an equality and not an inequality. Strictly, the consumer can afford anything with $p_x x + p_y y \leq m$. But “more is better” means unspent income buys another unit of something, which is preferred. So the chosen bundle always sits *on* the line, and we may write it as an equation.

What each term means. $3x$ is dollars spent on x — price per unit times units. Not a quantity. The equation balances dollars against dollars.

The separation being set up. The budget constraint contains prices and income and *no* preference information. The utility function contains preference information and *no* prices. That separation is what makes the comparative statics of Parts 5 to 7 possible: a price change moves one object and leaves the other exactly where it was.

Step 2 — plug the numbers in

Step 2 — plug the numbers in

Fix x , see what income is left, divide by $p_y = 2$.

x	0	8	16	20	24	32	40
spent on x : $3x$	0	24	48	60	72	96	120
left over: $120 - 3x$	120	96	72	60	48	24	0
$y = (120 - 3x)/2$	60	48	36	30	24	12	0

Every 4 units of x costs exactly 6 units of y — all the way along.
That constancy is why the line is straight.

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Slide 16

Working through it

One column worked in full. Take $x = 16$.

$$3 \times 16 = 48 \text{ spent on } x, \quad 120 - 48 = 72 \text{ left}, \quad 72/2 = 36 \text{ units of } y$$

Check: $3(16) + 2(36) = 48 + 72 = 120$. ✓

Why every column balances. By construction — y is defined as whatever the residual income buys. The table cannot fail this check, which is why it is worth doing on a row you did *not* construct: at $x = 24$, $3(24) + 2(24) = 72 + 48 = 120$. ✓

The constant rate. Between $x = 8$ and $x = 16$, y falls from 48 to 36 — 8 more x costs 12 of y , so 4 more x costs 6. Between $x = 24$ and $x = 32$, y falls from 24 to 12: again 8 for 12. The rate never changes, and a constant rate of exchange is exactly what a straight line is.

Contrast with the indifference curve. There, the rate at which y fell per unit of x *changed* as x rose — that was the flattening. Here it does not. One object curves and one is straight, and that difference is what produces a unique optimum in Part 3.

Step 3 — rearrange to plot**Step 3 — rearrange to plot**

Make y the subject:

$$3x + 2y = 120 \implies 2y = 120 - 3x \implies y = 60 - 1.5x$$

- **60** is m/p_y — all the income spent on y .
- **1.5** is p_x/p_y — how much y each unit of x costs.

Nothing economic has happened. It is the same constraint, written the other way round.

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Slide 17

Working through it

The general rearrangement.

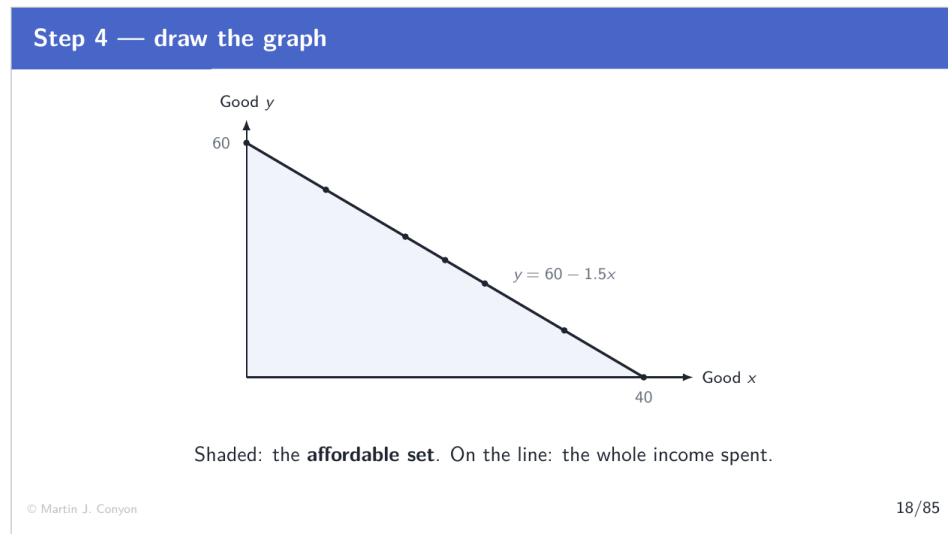
$$p_x x + p_y y = m \implies y = \frac{m}{p_y} - \frac{p_x}{p_y} x$$

so the vertical intercept is m/p_y and the slope is $-p_x/p_y$.

The numbers here. $m/p_y = 120/2 = 60$ and $p_x/p_y = 3/2 = 1.5$.

Why rearrange at all. Because of the axis convention: quantity of x horizontal, quantity of y vertical. To plot the constraint we need y as a function of x . This is the same step taken with a demand curve when price is made the subject, and for the same reason.

What to hold on to. The slope carries *only* the price ratio, and the intercept carries income and one price. So a change in p_x tilts the line and leaves the intercept alone; a change in income moves the line without tilting it. Part 5 uses the first of those and Part 7 uses both.

Step 4 — draw the graph

Slide 18

Working through it

The two intercepts, and where they come from.

$$\text{all income on } y : \frac{m}{p_y} = \frac{120}{2} = 60, \quad \text{all income on } x : \frac{m}{p_x} = \frac{120}{3} = 40$$

Both are computed in the figure from the declared macros, not typed as coordinates.

The affordable set. The shaded triangle is every bundle with $3x + 2y \leq 120$, together with the two axes. Its boundary is the budget line.

Why the consumer ends up on the boundary. A bundle strictly inside leaves income unspent. Spending it buys more of at least one good, and more is better. So the optimum is on the line — which is what lets Part 3 search along a line rather than over a whole region.

What is outside. Everything above the line is unaffordable at these prices and this income. Not undesirable — unaffordable. The consumer would happily take any of it.

The slope is a relative price

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The slope 1.5 is not a quantity of money. It is a **rate of exchange between goods**.

\$3 buys one unit of x .
 \$3 buys one and a half units of y .
So the price of x , measured in y , is 1.5.

To get one more x , the market makes you give up 1.5 of y . That is an opportunity cost.

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Working through it

The computation. With $p_x = \$3$ and $p_y = \$2$, one unit of x costs the same as $p_x/p_y = 3/2 = 1.5$ units of y . Dollars cancel: the ratio of two prices is a ratio of goods.

Why this reframing is worth the frame. It puts the budget constraint and the indifference curve into the same units. The indifference curve's slope is the rate at which the *consumer* will swap y for x ; the budget line's slope is the rate at which the *market* will. Two rates, same units, directly comparable — and Part 3 is nothing more than comparing them.

The connection to opportunity cost. The opportunity cost of a unit of x is what must be given up to get it. At these prices that is 1.5 units of y . It is the same idea used to read the slope of a production frontier, on the other side of the market.

A check on the direction. If x became dearer — say $p_x = \$6$ — the ratio rises to 3. Each x now costs 3 of y . The line gets steeper, which is exactly the pivot in Part 5.

3. Part 3: The Optimum

Which affordable bundle is best?

Which affordable bundle is best?

We now have two objects on the same pair of axes:

- a **budget line** — everything the consumer can afford;
- a **family of indifference curves** — how the consumer ranks bundles, higher curve meaning better.

The problem: find the affordable bundle on the highest curve it can reach.

The answer is on the line, never inside it. Money left unspent buys more of something, and more is better.

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Slide 23

Working through it

The problem, stated formally.

$$\max_{x,y} \sqrt{xy} \quad \text{subject to} \quad 3x + 2y = 120$$

Maximize one function, subject to a constraint that another function holds. That is the entire content of consumer theory.

Why the constraint is an equality. “More is better” rules out the interior of the affordable triangle, as Part 2 established. The search is one-dimensional: pick x , and the budget line determines y .

The three ways it will be solved. This part does it twice — by tabulating xy along the line, and by an argument about rates of exchange that requires no arithmetic at all. Part 4 does it a third time, with calculus, and gets a formula that works for any prices and any income.

Why bother with three. Because the numerical route shows *that* the answer is $(20, 30)$, the argument shows *why* it must be a tangency, and the calculus shows what happens when the numbers change.

Plug every affordable bundle in

Plug every affordable bundle in

Substitute $y = 60 - 1.5x$ into xy and read down the column.

x	8	12	16	20	24	28	32
$y = 60 - 1.5x$	48	42	36	30	24	18	12
xy	384	504	576	600	576	504	384

It climbs to $x = 20$ and falls away — symmetrically.

The best affordable bundle is (20, 30), and no calculus was used.

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Slide 24

Working through it

Why xy and not $\sqrt{xy}$. Squaring is strictly increasing on positive numbers, so it leaves the ranking untouched. Comparing 600 with 576 is easier than comparing 24.49 with 24.00, and the conclusion is identical.

Every column, checked.

$$\begin{aligned}
 x = 8 : y &= 60 - 12 = 48, & xy &= 384 \\
 x = 12 : y &= 60 - 18 = 42, & xy &= 504 \\
 x = 16 : y &= 60 - 24 = 36, & xy &= 576 \\
 x = 20 : y &= 60 - 30 = 30, & xy &= 600 \leftarrow \text{largest} \\
 x = 24 : y &= 60 - 36 = 24, & xy &= 576 \\
 x = 28 : y &= 60 - 42 = 18, & xy &= 504 \\
 x = 32 : y &= 60 - 48 = 12, & xy &= 384
 \end{aligned}$$

Why it is symmetric. Substituting gives $xy = x(60 - 1.5x) = 60x - 1.5x^2$, a downward parabola in x . Its two roots are $x = 0$ and $x = 40$, so its peak is halfway between them, at $x = 20$. The symmetry in the table is the symmetry of a parabola about its vertex.

What this settles and what it does not. It settles the answer for *these* prices and *this* income,

by exhaustion. It gives no way of answering the same question when p_x changes — which is why the argument on the next frames, and the formula in Part 4, are needed.

Why a non-tangency cannot be best — too much y

Why a non-tangency cannot be best — too much y

Take the affordable bundle (10, 45). Check: $3(10) + 2(45) = 30 + 90 = 120$. ✓

Consumer's rate: $MRS = y/x = 45/10 = 4.5$

Market's rate: $p_x/p_y = 1.5$

The consumer would hand over **4.5** units of y for one more x . The market charges **1.5**.

x is worth three times what it costs them. So they buy more x .

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Slide 25

Working through it

The bundle, checked against the budget. $3(10) + 2(45) = 30 + 90 = 120$. It is affordable and it lies exactly on the line.

The two rates.

$$MRS = \frac{y}{x} = \frac{45}{10} = 4.5 \quad \text{versus} \quad \frac{p_x}{p_y} = \frac{3}{2} = 1.5$$

Reading them. The first says: *at this bundle*, the consumer is willing to give up 4.5 of y to get one more x and be no worse off. The second says: the market will make that trade at 1.5. The consumer is willing to pay far more than the going rate, so the trade is worth making.

The gain, made concrete. Move from (10, 45) to (20, 30) along the budget line: xy goes from 450 to 600. Utility rises from $\sqrt{450} = 21.2$ to $\sqrt{600} = 24.5$.

The general principle. Whenever $MRS \neq p_x/p_y$ there is an unexploited trade along the budget line. A bundle with an unexploited gain cannot be the best one. So the optimum

must have the two rates equal — which is the tangency condition, arrived at without differentiating anything.

The same argument the other way — too much x

The same argument the other way — too much x

Now take (30, 15). Check: $3(30) + 2(15) = 90 + 30 = 120$. ✓

Consumer's rate: $MRS = 15/30 = \mathbf{0.5}$

Market's rate: $p_x/p_y = \mathbf{1.5}$

The consumer will give up only **0.5** of y for another x . The market demands **1.5**.

Now x is dear relative to its worth. So they sell x and buy y .

Both ends push toward the middle.

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Slide 26

Working through it

The bundle, checked. $3(30) + 2(15) = 90 + 30 = 120$. ✓

The two rates.

$$MRS = \frac{15}{30} = 0.5 \quad \text{versus} \quad \frac{p_x}{p_y} = 1.5$$

Reading them. Giving up one x releases \$3, which buys 1.5 of y . The consumer only needed 0.5 of y to be compensated for the lost x . So the trade leaves them strictly better off, and they take it.

The gain, made concrete. xy at (30, 15) is 450. At (20, 30) it is 600. Exactly the same number as the bundle in the previous frame — and that is the parabola's symmetry again: (10, 45) and (30, 15) are equidistant from the peak.

The pincer. From the left the consumer wants more x ; from the right, less. The only bundle from which they want to move in neither direction is the one where the two rates coincide. On a curve that flattens steadily and a line of constant slope, there is exactly one such

bundle.

The tangency condition

The tangency condition

$$\underbrace{MRS}_{\text{the consumer's rate}} = \underbrace{\frac{p_x}{p_y}}_{\text{the market's rate}} \quad \text{i.e.} \quad \frac{y}{x} = \frac{p_x}{p_y}$$

Here: $\frac{y}{x} = \frac{3}{2}$, and at (20, 30) that reads $\frac{30}{20} = 1.5$. ✓

The consumer's exchange rate and the market's agree. No trade is left that gains anything.

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Slide 27

Working through it

The condition, in both forms.

$$MRS = \frac{p_x}{p_y} \quad \Longleftrightarrow \quad \frac{y}{x} = \frac{p_x}{p_y}$$

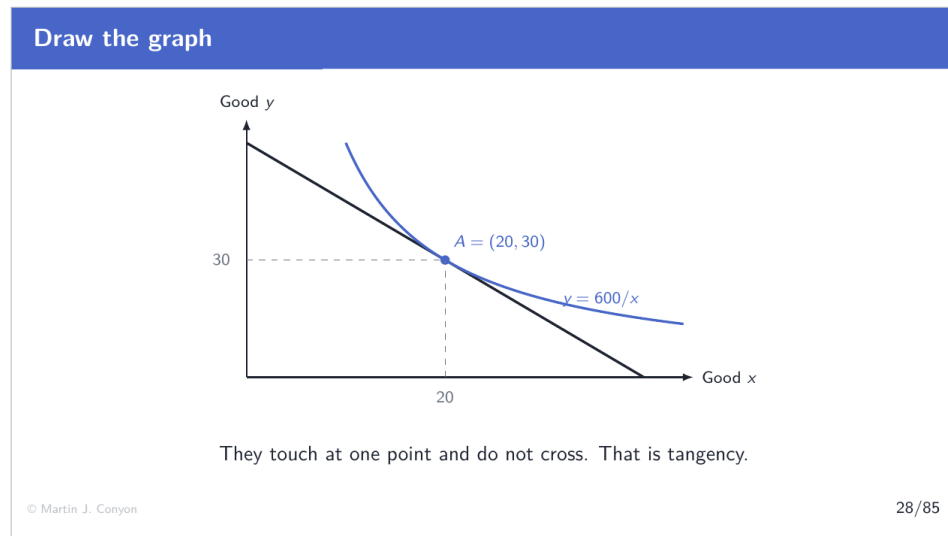
The second form is particular to $U = \sqrt{xy}$, where $MRS = y/x$. The first form is general.

Verified at the optimum. $y/x = 30/20 = 1.5$ and $p_x/p_y = 3/2 = 1.5$. The two are equal, and the bundle also satisfies $3(20) + 2(30) = 120$. Two conditions, two unknowns, one bundle.

Why “tangency”. Equal slopes at a shared point is what it means for a line to be tangent to a curve. The budget line touches the indifference curve and does not cross it.

What goes wrong if the curve is not convex. The argument used the fact that MRS falls steadily as x rises — that is what made the pincer work. For a curve that did not flatten steadily, a tangency could be a worst point rather than a best one. Our function is well behaved, and the tabulated column in this part is the direct evidence: the values rise to the tangency and fall after it.

Draw the graph



Slide 28

Working through it

How the figure is built. The budget line runs between the computed intercepts $(0, m/p_y)$ and $(m/p_x, 0)$. The curve is $y = k_A/x$ with $k_A = x_A y_A = 600$. The marked point is at (x_A, y_A) , computed from $x^* = m/(2p_x)$ and $y^* = m/(2p_y)$. If the algebra were wrong the dot would miss one of the two, and it does not.

Reading the picture. Three cases, and the figure shows the middle one:

- A curve **above** the line: unaffordable at every point.
- A curve **cutting** the line: parts of the line lie above it, so better affordable bundles exist. Not the optimum.
- A curve **touching** the line once: no affordable bundle is better. The optimum.

Why exactly one touching curve. The curves are nested and do not cross (Part 1), and they flatten steadily. Sliding outward from the origin, there is a first curve that reaches the line and a last that clears it; they are the same one.

4. Part 4: Solving It

Marginal utility, by differentiation

Marginal utility, by differentiation

Marginal utility. The extra utility from one more unit of a good, with the other held fixed. That is a partial derivative.

With $U = x^{1/2}y^{1/2}$:

$$MU_x = \frac{\partial U}{\partial x} = \frac{1}{2}x^{-1/2}y^{1/2} = \frac{1}{2}\sqrt{\frac{y}{x}}, \quad MU_y = \frac{\partial U}{\partial y} = \frac{1}{2}x^{1/2}y^{-1/2} = \frac{1}{2}\sqrt{\frac{x}{y}}$$

Each falls as you get more of that good.

Diminishing marginal utility — a consequence of the function, not an extra assumption.

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Working through it

The differentiation, step by step. Treat y as a constant:

$$\frac{\partial}{\partial x} (x^{1/2}y^{1/2}) = y^{1/2} \cdot \frac{1}{2}x^{-1/2} = \frac{1}{2} \frac{y^{1/2}}{x^{1/2}} = \frac{1}{2}\sqrt{\frac{y}{x}}$$

and symmetrically for y .

Why it is diminishing. $MU_x = \frac{1}{2}\sqrt{y/x}$ has x in the denominator, so raising x with y fixed lowers it. Numerically, with $y = 30$:

$$MU_x(10, 30) = \frac{1}{2}\sqrt{3} = 0.866, \quad MU_x(20, 30) = \frac{1}{2}\sqrt{1.5} = 0.612$$

$$MU_x(40, 30) = \frac{1}{2}\sqrt{0.75} = 0.433$$

The distinction worth keeping. Diminishing marginal *utility* is about one good at a time. Diminishing marginal rate of *substitution* is about the trade-off between two. They are different statements, and the second is what shapes the indifference curve.

The units warning. MU values are in utility per unit, and utility has no natural scale. Only their *ratio* means anything — which is the next frame.

The rate of substitution, proved

The rate of substitution, proved

Divide one marginal utility by the other:

$$MRS = \frac{MU_x}{MU_y} = \frac{\frac{1}{2}\sqrt{y/x}}{\frac{1}{2}\sqrt{x/y}} = \sqrt{\frac{y}{x}} \cdot \sqrt{\frac{y}{x}} = \frac{y}{x}$$

The $\frac{1}{2}$ s cancel. The exponents collapse. What is left depends only on the **ratio** y/x .

At (20, 30): $MRS = 30/20 = 1.5$ — which is what the chords in Part 1 were pointing at.

This is why Cobb–Douglas is tractable: the rate depends on the mix, never on the scale.

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Working through it

The algebra in full.

$$\frac{MU_x}{MU_y} = \frac{\frac{1}{2}x^{-1/2}y^{1/2}}{\frac{1}{2}x^{1/2}y^{-1/2}} = x^{-1/2-1/2}y^{1/2+1/2} = x^{-1}y = \frac{y}{x}$$

Why the ratio is the rate of substitution. Along an indifference curve $dU = MU_x dx + MU_y dy = 0$, so $-dy/dx = MU_x/MU_y$. The marginal utility of the good gained, divided by the marginal utility of the good given up, is the rate at which one substitutes for the other.

The confirmation of Part 1. There, chords on $y = 600/x$ gave -3 between $x = 10$ and $x = 20$, and -0.75 between $x = 20$ and $x = 40$. The exact point values from y/x are 6, 1.5 and 0.375 at $x = 10, 20, 40$. Each chord slope lies between the two point values at its ends, as an average rate must.

“Depends only on the ratio”. $MRS(20, 30) = 1.5$ and $MRS(40, 60) = 1.5$ — doubling both goods leaves the trade-off untouched. This property is what makes the closed forms in this part so clean, and it is the first thing Part 9 flags as special.

The Lagrangian

The Lagrangian

The problem:

$$\max_{x,y} x^{1/2}y^{1/2} \quad \text{subject to} \quad p_x x + p_y y = m$$

Build one function that carries both:

$$\mathcal{L}(x, y, \lambda) = x^{1/2}y^{1/2} + \lambda(m - p_x x - p_y y)$$

The bracket is **zero** at any bundle that spends the whole budget, so there $\mathcal{L} = U$.
Setting all three partial derivatives to zero solves the constrained problem.

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Working through it

What the method is for. Unconstrained maximization sets derivatives to zero. That fails here: raising x always raises U , so there is no interior maximum without the budget. The Lagrangian folds the constraint into the objective so the ordinary method works again.

Why the trick works. Write $g(x, y) = m - p_x x - p_y y$. On the budget line $g = 0$, so $\mathcal{L} = U + \lambda \cdot 0 = U$ there. The multiplier λ is an extra unknown, and the third condition $\partial \mathcal{L} / \partial \lambda = 0$ simply restates $g = 0$ — so the solution is guaranteed to satisfy the budget.

Three equations, three unknowns.

$$\frac{\partial \mathcal{L}}{\partial x} = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

solved for x , y and λ .

What λ turns out to mean. The next frame shows $MU_x = \lambda p_x$, so $\lambda = MU_x / p_x$: the extra utility bought by one more dollar. It is the marginal utility of income, and it is the same number whichever good you compute it from.

The two first-order conditions

The two first-order conditions

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2} \sqrt{\frac{y}{x}} - \lambda p_x = 0 \quad \implies \quad MU_x = \lambda p_x$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2} \sqrt{\frac{x}{y}} - \lambda p_y = 0 \quad \implies \quad MU_y = \lambda p_y$$

Each says: marginal utility equals λ times price.

Rearranged, $\frac{MU_x}{p_x} = \lambda = \frac{MU_y}{p_y}$ — the same utility per dollar in both goods.

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Working through it

The two conditions, written out.

$$\frac{1}{2} x^{-1/2} y^{1/2} = \lambda p_x, \quad \frac{1}{2} x^{1/2} y^{-1/2} = \lambda p_y$$

The equal-bang-per-buck reading. Divide each by its price:

$$\frac{MU_x}{p_x} = \lambda = \frac{MU_y}{p_y}$$

If this failed — say $MU_x/p_x > MU_y/p_y$ — then moving a dollar from y to x would raise utility, so the bundle was not optimal. The condition is the formal version of the trading argument in Part 3.

Checked at bundle $A = (20, 30)$.

$$MU_x = \frac{1}{2} \sqrt{30/20} = 0.6124, \quad MU_y = \frac{1}{2} \sqrt{20/30} = 0.4082$$

$$\frac{MU_x}{p_x} = \frac{0.6124}{3} = 0.2041, \quad \frac{MU_y}{p_y} = \frac{0.4082}{2} = 0.2041$$

Equal, so $\lambda = 0.2041$: one more dollar of income buys about 0.204 extra units of utility.

Divide them

Divide them

$$\frac{MU_x}{MU_y} = \frac{\lambda p_x}{\lambda p_y} \implies \frac{y}{x} = \frac{p_x}{p_y}$$

λ cancels — which is the whole point of dividing. It was never a quantity we cared about.

The tangency condition, argued in words in Part 3, falls straight out of the calculus.

The trading argument and the derivative give the same equation. That is the check: two routes, one answer.

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Working through it

The division.

$$\frac{MU_x}{MU_y} = \frac{\lambda p_x}{\lambda p_y} = \frac{p_x}{p_y}$$

and since $MU_x/MU_y = y/x$ for this utility function,

$$\frac{y}{x} = \frac{p_x}{p_y} \implies y = \frac{p_x}{p_y} x$$

Why λ disappears. It multiplies both prices identically, so it divides out. It carried no information about *which* bundle is best, only about how valuable an extra dollar would be.

The two routes compared.

- **Part 3, by argument.** If the consumer's rate of exchange differs from the market's, a gainful trade exists, so a best bundle must have them equal.
- **Part 4, by calculus.** Two first-order conditions, divided.

Same equation. The argument explains *why* it is true; the calculus makes it usable when the numbers change.

What is still missing. The tangency condition alone does not pin down a bundle — it is one equation in two unknowns, and it describes a whole ray from the origin. The budget constraint supplies the second equation.

Substitute into the budget — the closed forms

Substitute into the budget — the closed forms

Tangency gives $y = \frac{p_x}{p_y}x$. Put that into $p_x x + p_y y = m$:

$$p_x x + p_y \left(\frac{p_x}{p_y} x \right) = m \quad \Rightarrow \quad p_x x + p_x x = m \quad \Rightarrow \quad 2p_x x = m$$

$x^* = \frac{m}{2p_x}$

$y^* = \frac{m}{2p_y}$

Two formulas, good for *any* income and *any* prices.

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Working through it

The substitution, in full.

$$p_x x + p_y \left(\frac{p_x}{p_y} x \right) = m \Rightarrow p_x x + p_x x = m \Rightarrow 2p_x x = m \Rightarrow x^* = \frac{m}{2p_x}$$

Then from tangency,

$$y^* = \frac{p_x}{p_y} \cdot \frac{m}{2p_x} = \frac{m}{2p_y}$$

Read what is in each formula — and what is not. x^* contains m and p_x . It does *not* contain p_y . Likewise y^* contains m and p_y and not p_x . Each good's demand depends on income and its own price alone.

Why that is remarkable. It says a change in the price of x leaves the quantity of y exactly where it was. That is a strong claim about the world, and it is a property of this particular utility function rather than of consumers in general. Part 5 sees it happen and Part 9 says

what it costs.

The general Cobb–Douglas result. For $U = x^\alpha y^\beta$ with $\alpha + \beta = 1$, the optimum is $x^* = \alpha m / p_x$ and $y^* = \beta m / p_y$. With $\alpha = \beta = \frac{1}{2}$ that is the boxed pair.

Plug the numbers in — bundle A

Plug the numbers in — bundle A

$$x^* = \frac{120}{2(3)} = \frac{120}{6} = 20, \quad y^* = \frac{120}{2(2)} = \frac{120}{4} = 30$$

Check	Arithmetic	
Budget constraint	$3(20) + 2(30) = 60 + 60 = 120$	✓
Tangency	$y/x = 30/20 = 1.5$ and $p_x/p_y = 3/2 = 1.5$	✓

Both conditions hold exactly — and this is the bundle the table in Part 3 found by exhaustion.

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Working through it

The two computations.

$$x^* = \frac{m}{2p_x} = \frac{120}{6} = 20, \quad y^* = \frac{m}{2p_y} = \frac{120}{4} = 30$$

Both conditions, verified independently.

- **Budget:** $3(20) + 2(30) = 60 + 60 = 120$, exactly income.
- **Tangency:** $y/x = 30/20 = 1.5$ and $p_x/p_y = 3/2 = 1.5$.

Three routes, one answer. The table in Part 3 found $(20, 30)$ by computing xy at seven bundles. The trading argument showed the optimum must be a tangency. The calculus produced a formula that returns $(20, 30)$. Three methods agreeing is the strongest check

available, and it is worth noticing that the first required no theory and the third requires no arithmetic.

Utility at the optimum. $U = \sqrt{20 \times 30} = \sqrt{600} = 24.49 \dots$ The number itself is meaningless; what matters is that no affordable bundle gives a larger one.

Half the budget on each good — whatever the prices

Half the budget on each good — whatever the prices

Spending on x	$p_x x^* = 3 \times 20 = \60
Spending on y	$p_y y^* = 2 \times 30 = \60

And that is not the numbers being kind:

$$p_x x^* = p_x \cdot \frac{m}{2p_x} = \frac{m}{2}$$

The price cancels. **Whatever p_x is.**

Equal exponents put half the budget on each good and keep it there.

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Working through it

The cancellation.

$$p_x x^* = p_x \cdot \frac{m}{2p_x} = \frac{m}{2} = \$60 \quad \text{and} \quad p_y y^* = p_y \cdot \frac{m}{2p_y} = \frac{m}{2} = \$60$$

Neither expression contains a price. The *quantities* respond to prices; the *shares* do not.

Why this matters for what follows. Three results in this deck are direct consequences:

- The demand curve is $x^* = 60/p_x$, so expenditure is \$60 at every price — Part 6.
- That constant expenditure is exactly what **unit elasticity** means — Part 6.
- A change in p_x leaves spending on y untouched at \$60, so the quantity of y never moves — Part 5.

Where the property comes from. The exponents. For $U = x^\alpha y^\beta$ with $\alpha + \beta = 1$, the share spent on x is α . Ours are both $\frac{1}{2}$, hence the even split. It is a genuine restriction on behavior, and Part 9 treats it as one.

5. Part 5: A Price Change

The price of x doubles

The price of x doubles

	before	after
p_x	\$3	\$6
p_y	\$2	\$2
income m	\$120	\$120

Predict before solving: x is dearer, so the consumer buys *less* of it.

The question is how much less — and the model answers exactly.

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Working through it

What has changed and what has not. One number: p_x , from 3 to 6. Income, the price of y , and the utility function are all untouched. That isolation is the whole point of a comparative-statics exercise — whatever happens can be attributed to the one thing that moved.

Why the direction is predictable in advance. Two separate reasons, and Part 7 will insist on keeping them apart:

- x is now dearer *relative to* y , so the consumer swings toward y ;
- \$120 buys less than it did, so the consumer is poorer.

Both push x down here.

Why the magnitude is not predictable in advance. “Less” is a direction. Whether the answer is 19 or 10 depends on the shape of preferences, and that is precisely what the utility function encodes. Guessing is not available; solving is.

What to watch for. The quantity of y . Almost everyone expects it to move.

What moves, and what does not

What moves, and what does not

New budget: $6x + 2y = 120$, so $y = 60 - 3x$.

	before	after	why
y -intercept m/p_y	60	60	neither m nor p_y moved
x -intercept m/p_x	40	20	\$120 now buys half as many x
slope p_x/p_y	1.5	3	each x now costs 3 of y

The line **pivots** about the point on the vertical axis. It does not shift.

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Working through it

The three numbers, computed.

$$\frac{m}{p_y} = \frac{120}{2} = 60 \text{ (unchanged), } \frac{m}{p_x} = \frac{120}{6} = 20 \text{ (was 40), } \frac{p_x}{p_y} = \frac{6}{2} = 3 \text{ (was 1.5)}$$

Pivot versus shift — the distinction to hold on to.

- A change in a **price** tilts the line: one intercept moves, the other does not. The set of affordable bundles changes shape.
- A change in **income** moves the line parallel to itself: both intercepts move, the slope does not.

Part 7 uses exactly this difference to take the response apart: it introduces a parallel shift

to undo the income consequence of the pivot.

Why the vertical intercept is the anchor. It is the bundle where the consumer buys no x at all. Someone buying no x is unaffected by the price of x , so their position cannot move. Everything else on the line does.

The affordable set shrinks. Every bundle affordable after was affordable before, and some that were affordable before are not now. The consumer is unambiguously worse off.

Route one — the formula

Route one — the formula

$$x^* = \frac{m}{2p_x} = \frac{120}{12} = 10, \quad y^* = \frac{m}{2p_y} = \frac{120}{4} = 30$$

	before A	after B
x	20	10
y	30	30

The quantity of x halves. The quantity of y does not move at all.

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Working through it

The arithmetic.

$$x^* = \frac{120}{2 \times 6} = \frac{120}{12} = 10, \quad y^* = \frac{120}{2 \times 2} = \frac{120}{4} = 30$$

Why x halves exactly. $x^* = m/(2p_x)$ is inversely proportional to p_x . Double the price, halve the quantity — exactly, not approximately. This is the same property that makes the demand curve in Part 6 a rectangular hyperbola.

Why y does not move. $y^* = m/(2p_y)$ contains no p_x . The consumer's spending on y was \$60 before and is \$60 now, and at \$2 a unit that is 30 units either way.

Is that plausible? As a description of a real consumer, only sometimes. If x and y were coffee and tea, a doubling in the price of coffee would surely move tea. The model says it does not, and that is a limitation of Cobb–Douglas rather than a discovery about consumers. Part 9 states it plainly.

Bundle B. $(10, 30)$, with $k_B = 10 \times 30 = 300$ — a lower indifference curve than A 's 600, as it must be.

Route two — tangency and budget, solved directly

Route two — tangency and budget, solved directly

No formula. Two equations, two unknowns.

Tangency: $\frac{y}{x} = \frac{p_x}{p_y} = \frac{6}{2} = 3$, so $y = 3x$.

Budget: substitute it in —

$$6x + 2(3x) = 6x + 6x = 12x = 120 \quad \implies \quad x = 10, \quad y = 30$$

Budget check	$6(10) + 2(30) = 60 + 60 = 120 \checkmark$
Tangency check	$30/10 = 3$ and $6/2 = 3 \checkmark$

Both routes give $(10, 30)$.

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Working through it

Why two routes are worth the time. The formula is fast but opaque — it gives an answer with no visible reasoning. The direct route shows the two conditions doing their separate jobs: tangency fixes the *ratio* of the goods, the budget fixes the *scale*.

The solution, step by step.

$$y = 3x \implies 6x + 2(3x) = 120 \implies 12x = 120 \implies x = 10 \implies y = 3(10) = 30$$

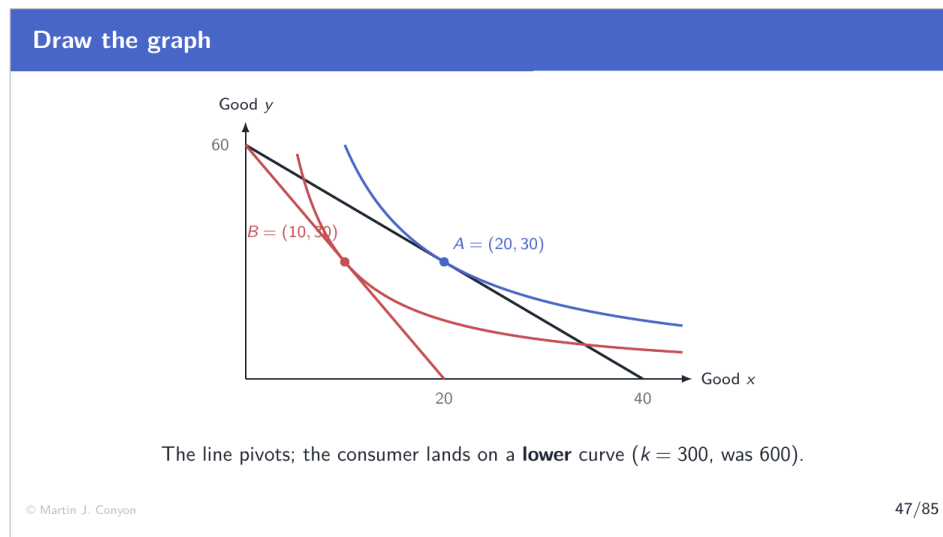
Both checks.

- Budget: $6(10) + 2(30) = 60 + 60 = 120$. $\checkmark$

- Tangency: $y/x = 30/10 = 3 = p_x/p_y$. ✓

What a disagreement would have meant. The two routes share no arithmetic at all — one divides, the other substitutes and solves. An error in either would show up as a mismatch. Agreement is not proof, but disagreement is certain proof of a mistake, and that asymmetry is why the habit is worth having.

Draw the graph



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Working through it

How the figure is built. Two budget lines from the computed intercepts: (0, 60) to (40, 0), and (0, 60) to (20, 0). Two indifference curves, $y = k_A/x$ with $k_A = 600$ and $y = k_B/x$ with $k_B = x_B y_B = 300$. Two marked points at the solved optima. Nothing is placed by eye.

The visual check. Each dot must sit on *both* its budget line and its indifference curve. That is four separate coincidences, and each one is a consequence of the algebra being right.

What the picture shows that the numbers do not. The two tangency points lie at the same height — the horizontal position moved and the vertical position did not. That is $y^* = m/(2p_y)$ made visible.

Utility before and after.

$$U_A = \sqrt{600} = 24.49 \dots, \quad U_B = \sqrt{300} = 17.32 \dots$$

The fall is real, and it is the reason Part 7 has to compensate the consumer before it can isolate the pure effect of the relative price.

Spending on y did not move

Spending on y did not move			
	before	after	
spent on x	$3(20) = \$60$	$6(10) = \$60$	unchanged
spent on y	$2(30) = \$60$	$2(30) = \$60$	unchanged
quantity of y	30	30	unchanged

The formula says why: $y^* = \frac{m}{2p_y}$ — and p_x appears nowhere in it.

With this utility function, a change in the price of x leaves the market for y untouched.

That is a strong property. Part 9 returns to it.

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Working through it

The four numbers.

$$3(20) = \$60, \quad 6(10) = \$60, \quad 2(30) = \$60, \quad 2(30) = \$60$$

Half the budget on each good before and after, exactly as Part 4 established.

The general statement. For this utility function, the cross-price effect is **zero**: $\partial y^* / \partial p_x = 0$. The two goods are neither substitutes nor complements in the usual sense — they are independent in demand.

Why a student should be suspicious of that. Most pairs of goods are not like this. A rise in the price of rail travel moves coach bookings; a rise in the price of printers moves cartridge sales. Any model that rules such effects out by construction cannot be used to study them.

The honest position. Cobb–Douglas is chosen for tractability, and it buys that tractability with a restriction on behavior. Knowing exactly which restriction, and being able to say it in one sentence, is the difference between using a model and being used by it.

6. Part 6: Deriving the Demand Curve

Do it for every price

Do it for every price

We solved the problem at $p_x = \$3$ and at $p_x = \$6$. There is no reason to stop.

$$x^* = \frac{m}{2p_x} \quad \text{with } m = 120 : \quad \boxed{x^* = \frac{60}{p_x}}$$

One equation. Every price at once. That is a demand function.

Nothing new is assumed here. It is the Part 4 result with the income written in.

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Working through it

Where the 60 comes from.

$$x^* = \frac{m}{2p_x} = \frac{120}{2p_x} = \frac{60}{p_x}$$

The 60 is $m/2$: half the budget, in dollars.

What kind of object this is. A **demand function** maps a price to the quantity a consumer chooses when doing the best they can at that price. It is a solved optimization problem, one answer per price.

Why this is the point of the whole session. A demand curve has so far been something written down — $Q^d = 70 - 10P$, say — with a downward slope asserted because it seemed sensible. Here it is a *conclusion*. We assumed preferences and a budget, we maximized, and a demand curve came out. Its slope is not a supposition; it is a result, and we know the

exact equation.

One caution about scope. This is *one* consumer's demand curve. A market demand curve is what you get by adding such curves horizontally across consumers, which is a separate step.

Step 2 — plug the numbers in

Step 2 — plug the numbers in

p_x	\$1	\$2	\$3	\$4	\$5	\$6
$x^* = 60/p_x$	60	30	20	15	12	10
spent: $p_x x^*$	\$60	\$60	\$60	\$60	\$60	\$60

Every entry in the middle row is 60 divided by the price above it — check any of them.

And the bottom row never changes. Hold on to that.

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Working through it

Every column, checked.

$$\frac{60}{1} = 60, \quad \frac{60}{2} = 30, \quad \frac{60}{3} = 20, \quad \frac{60}{4} = 15, \quad \frac{60}{5} = 12, \quad \frac{60}{6} = 10$$

And the expenditure row.

$$1(60) = 60, \quad 2(30) = 60, \quad 3(20) = 60, \quad 4(15) = 60, \quad 5(12) = 60, \quad 6(10) = 60$$

All \$60 — half the \$120 income, as Part 4 proved it must be.

Two of these columns are already familiar. At $p_x = \$3$ the answer is 20, which is bundle A. At $p_x = \$6$ it is 10, which is bundle B. The table contains the two cases solved the long way, and four more solved for free.

The law of demand, visible. Quantity falls as price rises, in every column. Here that is not an assumption imported from elsewhere — it is what $60/p_x$ does.

Step 3 — rearrange to plot

Step 3 — rearrange to plot

Economists put price on the *vertical* axis, so make p_x the subject:

$$x^* = \frac{60}{p_x} \quad \Longleftrightarrow \quad p_x = \frac{60}{x}$$

The same relationship, written the other way round.

Both forms say $p_x \cdot x = 60$. It is a **rectangular hyperbola** — the same curve family as an indifference curve, on different axes and for a completely different reason.

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Working through it

The rearrangement.

$$x = \frac{60}{p_x} \implies x p_x = 60 \implies p_x = \frac{60}{x}$$

One relationship, two ways of writing it. Nothing economic happens in this step; it is a concession to the axis convention inherited from Alfred Marshall.

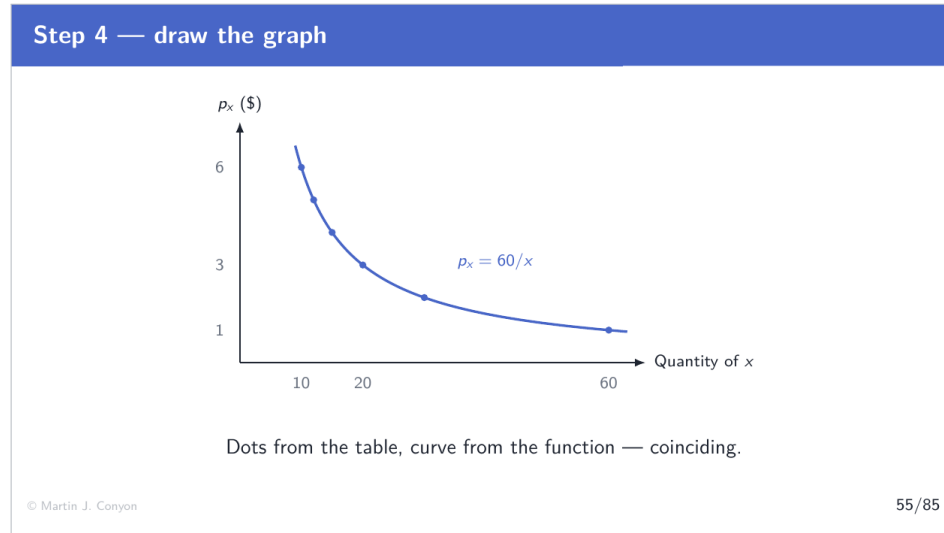
A coincidence worth naming so it does not confuse. Indifference curves here are $y = k/x$ and this demand curve is $p = 60/x$. Both are rectangular hyperbolas, and they have *nothing* to do with each other:

- the indifference curve plots y against x , two goods, at fixed utility;
- the demand curve plots price against quantity, one good, at the consumer's optimum.

The shared shape is an artefact of the utility function, not a connection between the two objects.

What the constant 60 is. Expenditure on x in dollars. The curve is the locus of price–quantity pairs whose product is the consumer’s spending.

Step 4 — draw the graph



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Working through it

How the figure is built. The curve is plot of $p = \backslash\text{Dconst}/x$ with $\backslash\text{Dconst} = m/2 = 60$. The six dots are placed by a loop that computes $x = 60/p$ for $p = 1, \dots, 6$. Both come from the same declared income.

Reading the shape. Steep at the top left, flat at the bottom right. At high prices a small price change moves quantity a little; at low prices it moves quantity a lot — in *absolute* terms. In percentage terms, as the next frame shows, the responsiveness is identical everywhere.

The asymptotes. $p = 60/x$ approaches both axes without reaching either. No price is high enough to drive purchases to zero, and no price is low enough to produce unbounded demand. That is a property of this utility function; real demand curves usually hit the price axis at a choke price.

The check that matters. The point (20, 3) is bundle A ’s x -quantity at A ’s price, and (10, 6) is bundle B ’s. The two problems solved at length in Parts 4 and 5 are two dots on this curve.

What has just happened

What has just happened

Before: a demand curve was written down, and assumed to slope downward because that seemed sensible.

Here: nothing about demand was assumed at all.

- preferences $U = \sqrt{xy}$
- a budget $p_x x + p_y y = m$
- do the best you can

Out came: $x^* = 60/p_x$.

The demand curve is a *result*, and we know its exact equation.

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Working through it

The logical structure, which is the point of the frame.

preferences + budget + optimization $\implies$ demand curve

The arrow runs one way. Nothing about the demand curve was fed in.

What this buys. Three things a written-down demand curve cannot give:

- **An explanation of the slope.** It falls because of how a constrained optimum responds to a price, not because it seemed reasonable.
- **Exact comparative statics.** Change income and the curve moves in a way you can compute: $x^* = m/(2p_x)$.
- **A testable structure.** The model predicts constant expenditure shares, which is a claim that can be checked against data — and often found wanting.

The honest limit. What has been derived is *this* consumer's demand curve under *this* utility function. The general result — that a compensated demand curve must slope down whatever preferences look like — needs the argument in Part 8.

Expenditure is \$60 at every price**Expenditure is \$60 at every price**

Price $\times$ quantity, from the table: $1(60) = 2(30) = \dots = 6(10) = \60 .

Constant expenditure is exactly what **unit elastic** means. Check it with the midpoint method, p_x from \$3 to \$6:

$$\% \Delta Q = \frac{10 - 20}{15} = -\frac{2}{3}, \quad \% \Delta P = \frac{6 - 3}{4.5} = +\frac{2}{3}$$

$$\text{elasticity} = \frac{-2/3}{+2/3} = -1$$

Exactly minus one — and it comes out at minus one for every pair of prices on this curve.

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Working through it

The midpoint computation in full. Between $(p, x) = (3, 20)$ and $(6, 10)$:

$$\% \Delta Q = \frac{10 - 20}{(10 + 20)/2} = \frac{-10}{15} = -0.6\bar{6}, \quad \% \Delta P = \frac{6 - 3}{(6 + 3)/2} = \frac{3}{4.5} = +0.6\bar{6}$$

$$E_d = \frac{-0.6\bar{6}}{+0.6\bar{6}} = -1$$

Why it is -1 for any pair, not just this one. With $x = 60/p$, take two prices p_1, p_2 :

$$E_d = \frac{(x_2 - x_1)(p_1 + p_2)}{(p_2 - p_1)(x_1 + x_2)} = \frac{60 \frac{p_1 - p_2}{p_1 p_2} (p_1 + p_2)}{(p_2 - p_1) \cdot 60 \frac{p_1 + p_2}{p_1 p_2}} = \frac{p_1 - p_2}{p_2 - p_1} = -1$$

Everything cancels. Try \$5 $\rightarrow$ \$6: quantity 12 to 10, giving $(-2/11)/(2/11) = -1$. Try \$1 $\rightarrow$ \$2: 60 to 30, giving $(-30/45)/(1/1.5) = -1$.

The chain back to Part 4. Equal exponents $\Rightarrow$ half the budget on each good $\Rightarrow$ expenditure on x constant $\Rightarrow$ unit elastic demand. Each link is exact, and the last one is the standard result that a fall in price raises quantity by exactly enough to leave revenue unchanged.

7. Part 7: Income and Substitution Effects

The fall from 20 to 10 is two things at once

The fall from 20 to 10 is two things at once

When p_x doubled, x fell from 20 to 10. **Two separate causes.**

1. x became dearer *relative to* y , so the consumer swings toward y .
A change in the rate of exchange.
2. The consumer became *poorer* — \$120 no longer buys what it did.
A change in what is affordable at all.

To separate them, remove the second one — then whatever is left is the first.

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Working through it

Why the two must be separated. They are different phenomena with different properties. The relative-price effect can be signed in advance — Part 8 proves it is never positive. The income effect cannot: it depends on whether the good is one people buy more of, or less of, as they get richer.

The names.

- **Substitution effect:** the change in x caused by the change in the relative price, holding purchasing power fixed.
- **Income effect:** the change in x caused by the change in purchasing power, holding relative prices at their new level.

The strategy. Invent an intermediate bundle: what the consumer would choose at the *new* prices if they had been given enough income to undo the loss. Then

$$\underbrace{A \rightarrow C}_{\text{prices change, purchasing power held}} \quad \underbrace{C \rightarrow B}_{\text{purchasing power removed, prices held}}$$

and the two steps together are exactly $A \rightarrow B$.

What “holding purchasing power fixed” means. It needs a definition, and there is more than one. The next frame gives the one used here.

Compensate: keep the old bundle just affordable

Compensate: keep the old bundle just affordable

Slutsky compensation. Give the consumer exactly enough income that the *original bundle* is affordable at the *new* prices.

Cost bundle $A = (20, 30)$ at the new prices:

$$m' = p_x^{\text{new}}x_A + p_y y_A = 6(20) + 2(30) = 120 + 60 = \textbf{\$180}$$

At \$180 the consumer *could* buy exactly what they bought before. So nothing has been taken away.

Only the relative price has changed.

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Working through it

The computation.

$$m' = p_x^{\text{new}}x_A + p_y y_A = 6(20) + 2(30) = 120 + 60 = 180$$

The compensation is $180 - 120 = \$60$ of extra income.

What the definition is doing. It fixes a benchmark for “the same purchasing power”: the consumer can still afford the exact basket they bought before. Whatever they do instead of buying it, they do because the relative price changed, not because they are poorer.

Why the old bundle is still just affordable. By construction, m' is precisely its cost. It sits on the compensated budget line, not inside it. That fact is used twice: once in the figure, where the compensated line visibly passes through A , and once in Part 8, where it is the first of the two facts the proof needs.

The alternative convention. Hicks compensation keeps *utility* unchanged rather than the old bundle affordable. It hands over slightly less money. Both are in use; Part 9 says what turns on the difference.

The compensated bundle C

The compensated bundle C

Solve again — new prices, compensated income $m' = \$180$:

$$x_C = \frac{m'}{2p_x} = \frac{180}{12} = \mathbf{15}, \quad y_C = \frac{m'}{2p_y} = \frac{180}{4} = \mathbf{45}$$

Budget check	$6(15) + 2(45) = 90 + 90 = 180 \checkmark$
Tangency check	$y/x = 45/15 = 3 \quad \text{and} \quad p_x/p_y = 6/2 = 3 \checkmark$

The compensated budget line is $y = 90 - 3x$: same slope as the new line,
further out.

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Working through it

The two computations.

$$x_C = \frac{180}{2(6)} = \frac{180}{12} = 15, \quad y_C = \frac{180}{2(2)} = \frac{180}{4} = 45$$

Both conditions, verified.

- Budget: $6(15) + 2(45) = 90 + 90 = 180$, exactly m' . $\checkmark$
- Tangency: $45/15 = 3$ and $6/2 = 3$. $\checkmark$

Half the compensated income on each good, \$90 and \$90 — the equal-shares property again.

The compensated budget line. $6x + 2y = 180$, so $y = 90 - 3x$. Its slope is -3 , the same as the new line's, because it faces the *same* prices. Its intercepts are $180/6 = 30$ and $180/2 = 90$, both further out.

The check that it passes through A . Substitute $x = 20$: $y = 90 - 3(20) = 90 - 60 = 30$, which is y_A . ✓ The old bundle sits exactly on the compensated line, which is what the compensation was built to achieve.

Note what C is not. It is not a bundle anyone was observed buying. It is a constructed counterfactual: what this consumer *would* choose at the new prices with purchasing power restored.

The split

The split

$$\underbrace{20 \rightarrow 15}_{\text{substitution: } -5} \quad \underbrace{15 \rightarrow 10}_{\text{income: } -5} \quad \underbrace{20 \rightarrow 10}_{\text{total: } -10}$$

- $A \rightarrow C$: relative price changes, purchasing power held fixed.
- $C \rightarrow B$: the compensation is taken away again, prices unchanged.

-5 and -5 sum to -10, the whole fall. They must.

That the two are *equal* here is a property of this utility function, not a law.

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Working through it

The decomposition, with every number.

$$\text{substitution} = x_C - x_A = 15 - 20 = -5$$

$$\text{income} = x_B - x_C = 10 - 15 = -5$$

$$\text{total} = x_B - x_A = 10 - 20 = -10 \quad \text{and} \quad -5 + (-5) = -10 \quad \checkmark$$

Why the sum is guaranteed. The middle term cancels: $(x_C - x_A) + (x_B - x_C) = x_B - x_A$. The identity holds whatever the numbers are, so it is a check on arithmetic rather than on economics — but it is worth doing, because it catches a mis-solved C immediately.

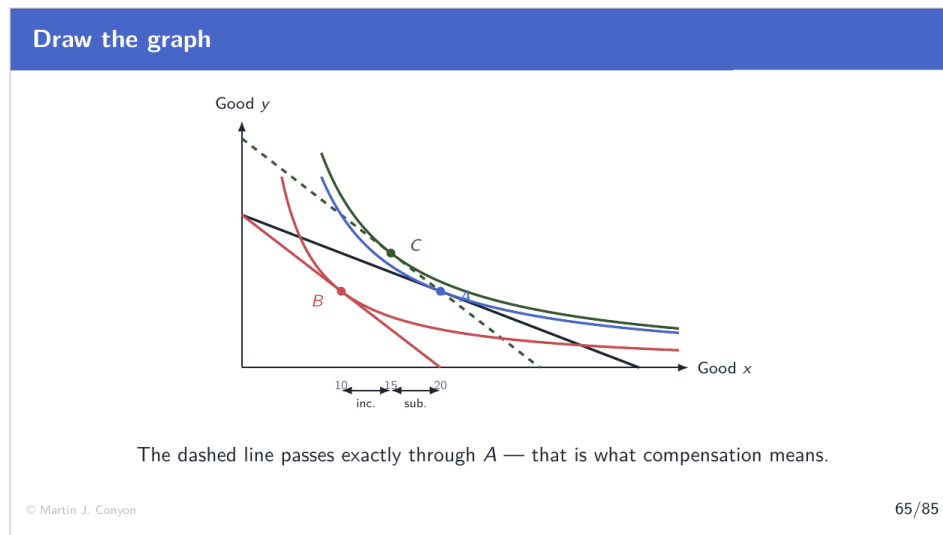
On the two effects being equal. It is not a coincidence and it is not general. For $U = \sqrt{xy}$ the algebra gives

$$\text{substitution} = \frac{m}{4p'_x} - \frac{m}{4p_x} = \text{income}$$

so the two are *always* equal for this utility function with these equal exponents. Change the utility function and they differ, usually substantially. The right statement is: equal here because of Cobb–Douglas, not equal in general.

Signs. Both are negative here. The substitution effect is *never* positive — Part 8 proves it. The income effect could have been positive, and would be for a good people buy less of as they get richer.

Draw the graph



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Working through it

How the figure is built. Three budget lines from computed intercept pairs: $(0, 60)$ – $(40, 0)$, $(0, 90)$ – $(30, 0)$ dashed, and $(0, 60)$ – $(20, 0)$. Three indifference curves, $y = k/x$ with $k_A = 600$, $k_C = 675$, $k_B = 300$. Three points at the solved optima. No coordinate is typed by hand.

What to look for, in order.

1. The dashed line passes through A — check $90 - 3(20) = 30$. ✓

2. The dashed line is *parallel* to the new line: both have slope -3 , because both face the new prices.
3. Each of the three dots lies on its own budget line and its own indifference curve.
4. C lies on a curve *outside* A 's.

The two effects, read off the horizontal axis. $20 \rightarrow 15$ is the substitution effect; $15 \rightarrow 10$ is the income effect. Both are movements to the left.

Why C is between A and B here. Both effects have the same sign, so they accumulate. For a good with a positive income effect the intermediate bundle would lie outside the two, and the picture would look quite different.

The compensation was generous

The compensation was generous

Which indifference curve does each bundle sit on?

$$k_A = 20(30) = 600, \quad k_C = 15(45) = \mathbf{675}, \quad k_B = 10(30) = 300$$

$675 > 600$: C is on a *higher* curve than A .

Keeping the old *bundle* affordable is more generous than keeping *utility* unchanged — because the consumer does not buy the old bundle. They buy something better.

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Working through it

The three constants.

$$k_A = 20 \times 30 = 600, \quad k_C = 15 \times 45 = 675, \quad k_B = 10 \times 30 = 300$$

In utility terms, $\sqrt{600} = 24.49$, $\sqrt{675} = 25.98$, $\sqrt{300} = 17.32$.

Why C beats A. The compensated income makes A affordable. But A is no longer the *best* affordable bundle at the new prices — its tangency condition fails, since $y_A/x_A = 1.5$ while $p_x/p_y = 3$. So the consumer re-optimizes, moves along the compensated line, and ends up strictly better off. Giving someone enough money to buy their old basket, after prices have tilted, leaves them slightly ahead.

The alternative. Hicks compensation asks: how much income keeps *utility* at $\sqrt{600}$? That is less than \$180. Solving it gives $m'' = 2\sqrt{k_A p_x p_y} = 2\sqrt{600 \times 12} = 169.7$, so about \$170 rather than \$180.

Does the difference matter? For the sign of the substitution effect, no — both conventions give a non-positive one, and Part 8's proof uses Slutsky. For magnitudes, slightly. Part 9 lists it as a limitation and leaves it there.

8. Part 8: Why the Substitution Effect Is Never Positive

The claim

The claim

Raise the price of a good and hold purchasing power fixed by compensating.
The consumer never buys *more* of it.

The proof uses:

- no calculus;
- no assumption about the shape of preferences;
- nothing about this particular utility function.

Only that the consumer **chooses consistently**.

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Working through it

The statement, precisely. Let A be chosen at prices (p_x, p_y) with income m . Raise p_x to $p'_x > p_x$, and compensate income to $m' = p'_x x_A + p_y y_A$. Let C be chosen at (p'_x, p_y) with

m' . Then

$$x_C \leq x_A$$

Why it deserves a proof rather than an assertion. Everything else in this deck is true of *Cobb–Douglas*. This is true of *any* consumer whose choices are consistent, whatever their preferences look like. It is one of the few genuinely general results in consumer theory, and it is what underwrites the claim that demand curves slope downward.

The method. Proof by contradiction, using **revealed preference** — the principle that a chosen bundle is at least as good as any bundle that was available and not chosen. No utility function need exist at all.

What the result does not say. It concerns *compensated* demand. Ordinary demand contains an income effect too, and the last frame of this part says what that allows.

The two facts we already have

The two facts we already have

Fact 1. Under the compensated budget, the old bundle A is exactly affordable.
That is what the compensation was constructed to do: $m' = p'_x x_A + p_y y_A$ is precisely the cost of A at the new prices.

Fact 2. Facing that budget, the consumer chose C — not A .
 A was available and was passed over.

So C is at least as good as A .

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Working through it

Fact 1, verified with the numbers. $m' = 180$, and the cost of A at the new prices is $6(20) + 2(30) = 180$. Equal, so A lies exactly on the compensated budget line. In the Part 7 figure, the dashed line visibly runs through A .

Fact 2, and what it licenses. The consumer, able to buy A , bought C instead. The revealed-preference principle says: a bundle chosen when another was affordable is at least as good as that other. So $C \succeq A$.

Why this is weaker than it sounds — and enough. It does not say C is *much* better, or that the consumer is happy, or that any utility number exists. It says only that the ranking cannot run the other way. That is all the contradiction needs.

The one assumption in play. Consistency. A consumer who chose C over an available A here, and A over an available C elsewhere, is not choosing consistently — and that is exactly the contradiction the next two frames construct.

Suppose the opposite

Suppose the opposite

Suppose the result is false. Suppose the compensated bundle C contains more x than A did:

$$x_C > x_A$$

We follow that supposition until it breaks.

Nothing else changes. C is still on the compensated budget line, and A is still just affordable on it.

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Working through it

The structure of the argument. Proof by contradiction. Assume the negation of what you want, derive something impossible, conclude the negation was false. The same shape as the “indifference curves cannot cross” argument in Part 1, and for the same underlying reason: a consistent ranking cannot contradict itself.

What is being supposed. That a rise in the price of x , with purchasing power restored, leads the consumer to buy *more* x . That would be an upward-sloping compensated demand

curve.

Why it is not obviously absurd. Nothing so far rules it out. Preferences are almost unrestricted — three weak assumptions and no shape requirement. It is entirely reasonable to ask whether some strange preference ordering might produce it. The next frame shows none can.

What stays fixed under the supposition.

- $p'_x x_C + p_y y_C = m'$ (C is on the compensated line)
- $p'_x x_A + p_y y_A = m'$ (A is on it too, by construction)

Those two equalities are the whole engine of the next step.

Cost the two bundles at the *old* prices

Cost the two bundles at the *old* prices

Both cost m' at the new prices, so extra spending on x must be matched by less on y . Subtracting the two budget identities gives:

$$\underbrace{(p_x x_C + p_y y_C)}_{\text{cost of } C, \text{ old prices}} - \underbrace{(p_x x_A + p_y y_A)}_{\text{cost of } A, \text{ old prices}} = (p_x - p'_x)(x_C - x_A)$$

Here $p_x - p'_x = 3 - 6 = -3$.

If $x_C > x_A$, the product is **negative**: C would have been strictly cheaper than A at the old prices.

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Working through it

The algebra, in full. Both bundles satisfy the compensated budget:

$$p'_x x_C + p_y y_C = m' = p'_x x_A + p_y y_A$$

so

$$p_y(y_C - y_A) = -p'_x(x_C - x_A)$$

Now the difference in cost at the *old* prices:

$$\begin{aligned}(p_x x_C + p_y y_C) - (p_x x_A + p_y y_A) &= p_x(x_C - x_A) + p_y(y_C - y_A) \\ &= p_x(x_C - x_A) - p'_x(x_C - x_A) \\ &= (p_x - p'_x)(x_C - x_A)\end{aligned}$$

The sign. $p_x - p'_x = 3 - 6 = -3$, negative because the price rose. So if $x_C - x_A > 0$ the whole expression is negative: C costs *less* than A at the old prices. Since A cost exactly m , bundle C would have cost less than m — strictly affordable, with money left over.

The check on the real numbers. Here $x_C = 15$ and $x_A = 20$, so $(3-6)(15-20) = (-3)(-5) = +15$: C costs \$15 *more* at the old prices. Directly: $3(15) + 2(45) = 45 + 90 = 135$, against A 's \$120. ✓ No contradiction arises — because the substitution effect was negative, exactly as the theorem requires.

The contradiction

The contradiction

At the old prices: C was affordable, with money to spare — and the consumer chose A .

$\Rightarrow$ A is preferred to C

Under the compensated budget: A was affordable — and the consumer chose C .

$\Rightarrow$ C is preferred to A

Both cannot hold. The supposition fails, so $x_C \leq x_A$ always.

The compensated demand curve slopes down — for *any* consistent consumer.

Working through it

The two revealed preferences, side by side.

Situation	Affordable	Chosen
Old prices, income m	both A and C	$A \Rightarrow A \succeq C$
New prices, income m'	both A and C	$C \Rightarrow C \succeq A$

With C strictly cheaper in the first row, the first preference is strict, and strict-plus-weak in opposite directions is impossible.

What has been established. $x_C \leq x_A$ whenever $p'_x > p_x$ and income is Slutsky-compensated. Equivalently: the compensated demand curve is never upward sloping.

What was used. Two budget identities and the principle that a chosen bundle is at least as good as an affordable rejected one. That is all. No differentiability, no convexity, no utility function.

Why economists care so much about this one result. It is the closest thing consumer theory has to a law. Almost everything else in the subject is a consequence of assumptions that could reasonably be doubted; this follows from consistency alone.

Ordinary demand can still slope up

Ordinary demand can still slope up

The theorem is about **compensated** demand. Ordinary demand contains the income effect too:

$$\underbrace{\text{total}}_{\text{ordinary demand}} = \underbrace{\text{substitution}}_{\text{never positive}} + \underbrace{\text{income}}_{\text{either sign}}$$

If the income effect is positive and *large enough* to outweigh the substitution effect, ordinary demand slopes **upward**.

Such a good is called a **Giffen good**.

They are rare, and hard to find convincingly in real data.

Working through it

The decomposition, as a statement about signs.

$$\underbrace{\frac{\Delta x}{\Delta p_x}}_{\text{ordinary}} = \underbrace{\text{substitution}}_{\leq 0 \text{ always}} + \underbrace{\text{income}}_{\text{sign varies}}$$

When the income effect is positive. A price rise makes the consumer poorer. If they respond to being poorer by buying *more* of the good — an inferior good — then the income effect pushes quantity up. For it to overturn the substitution effect it must be both positive and large, which requires the good to take a substantial share of the budget.

Why that combination is rare. A good has to be inferior *and* a large part of spending. Most inferior goods are small items; most large budget items are not inferior.

The status of the evidence. The classic story — Irish potatoes during the famine — is disputed on the historical record. Jensen and Miller (2008) report evidence from a subsidy experiment on staple foods in Hunan, China, and it remains one of very few credible cases.

The honest summary. Giffen goods are a logical possibility that the theory correctly permits, not a common feature of markets. Naming one frame for them is the right amount of attention.

9. Part 9: What the Model Leaves Out

Cobb–Douglas is a special case

Cobb–Douglas is a special case

Our utility function has a name — **Cobb–Douglas** — and it is convenient rather than general.

- Constant expenditure shares: always half on each good.
- Therefore unit-elastic demand: always exactly -1 .
- Therefore **no cross-price effect**: p_x can do what it likes and the quantity of y never moves.

So the two goods cannot be substitutes or complements in any interesting way.

Working through it

The three restrictions, and where each was visible.

- **Constant shares** — Part 4: $p_x x^* = m/2$ whatever p_x is.
- **Unit elasticity** — Part 6: every midpoint calculation gave exactly -1 , for every pair of prices.
- **Zero cross-price effect** — Part 5: p_x doubled and the quantity of y stayed at 30.

These are not three separate assumptions. They are one property seen from three angles, and that property is the equal exponents.

What is lost. Substitution and complementarity between goods — coffee and tea, printers and cartridges, cars and fuel. A model that sets those effects to zero by construction cannot be used to study them, and they are often the whole question.

What is kept. Everything in Parts 1, 2, 3, and 8. The apparatus — budget line, indifference curves, tangency, revealed preference — carries over to any preferences at all. It is the clean *numbers* that are special, not the method.

The general form. $U = x^\alpha y^\beta$ gives shares α and β ; richer functions such as CES allow genuine substitution effects between goods at the cost of much messier algebra.

Preferences are given, stable and consistent

Preferences are given, stable and consistent

The model takes preferences as **data**. It says nothing about:

- where they come from;
- whether advertising moves them;
- whether they change with age, mood, or company;
- whether the consumer knows what is available, or what things cost.

That is not a small thing to assume. The whole apparatus rests on it.

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Working through it

Why the assumption is load-bearing. Every comparative-statics exercise in the deck compares a before and an after. If preferences moved in between, the difference cannot be attributed to the price. The utility function has to be the *same* function at both prices for Parts 5 to 8 to mean anything.

What consistency in particular buys. It is the one assumption Part 8's proof uses, and it is the one most directly testable: observe a consumer's choices at several price-income combinations and check whether any pair contradicts. That test has a name — the weak axiom of revealed preference — and real consumers fail it sometimes.

The behavioral challenge. A body of evidence documents preferences that depend on how a choice is framed, on what is presented as the default, and on what the consumer happens to own already. None of that fits inside a fixed $U(x, y)$.

The defensible position. The model is a good first approximation for stable, repeated, low-stakes purchases, and a poor one for unfamiliar or emotionally loaded decisions. Knowing which is which is the skill.

Two goods

Two goods

The page has two dimensions, so the picture does.

But **nothing in the logic** depends on it. The tangency condition and the closed forms extend to any number of goods:

$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = \dots = \frac{MU_n}{p_n}$$

What is lost with more goods is the graph, not the argument.

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Working through it

The general problem.

$$\max_{x_1, \dots, x_n} U(x_1, \dots, x_n) \quad \text{subject to} \quad \sum_{i=1}^n p_i x_i = m$$

The Lagrangian carries $n + 1$ first-order conditions, one per good plus the constraint. Each gives $MU_i = \lambda p_i$, so dividing any pair gives $MU_i/MU_j = p_i/p_j$ — the same tangency condition, between any two goods.

What survives and what does not.

- **Survives:** the optimality condition, the Cobb–Douglas closed forms $x_i^* = \alpha_i m/p_i$, the decomposition into income and substitution effects, and the Part 8 proof, which never used the number of goods.
- **Does not survive:** the picture. Three goods need three axes; more need none we can draw.

The practical dodge. Put the good of interest on one axis and “spending on everything else” on the other. That is a composite commodity, and it is legitimate when the prices of the other goods move together.

Slutsky, not Hicks

Slutsky, not Hicks

We compensated to keep the *old bundle affordable*. That is **Slutsky**.

The alternative keeps *utility unchanged*. That is **Hicks**, and it hands over less money — about \$170 here rather than \$180, because we saw that \$180 left the consumer slightly better off.

Both give a substitution effect that is **never positive**. The magnitudes differ slightly.

A model whose limits you cannot state is a model you do not understand.

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Working through it

The two definitions, side by side.

	holds fixed	compensated income here
Slutsky	the old <i>bundle</i> is affordable	\$180
Hicks	the old <i>utility</i> is attained	about \$170

Why Slutsky gives more. At the new prices the old bundle is no longer optimal, so a consumer handed enough to buy it will re-optimize and do better. Part 7 saw this directly: $k_C = 675$ against $k_A = 600$.

The Hicks figure, for completeness. Keeping $U = \sqrt{600}$ at prices $(6, 2)$ requires $m'' = 2\sqrt{k_A p_x p_y} = 2\sqrt{600 \times 12} \approx 169.7$.

Why Slutsky is used here. It is *observable*. The old bundle and the new prices are both data; the old utility level is not. That is also why the Part 8 proof works with no utility function at all — it needs only bundles and prices.

What turns on the choice. The sign of the substitution effect: nothing, both give non-positive. The magnitude: a little. Which one a piece of empirical work should use: whichever the available data can actually construct.